Editorial note - portfolio edition (August 2026)

This paper was written in July 2025 as a capstone submission for the NUS Generative AI programme. It has not been peer reviewed or published in any journal, and it is presented here as a student capstone artefact rather than as a peer-reviewed research contribution.

Two changes have been made to the original submission before publishing it in this portfolio:

1. Eight academic citations have been removed. On review in August 2026, each could not be located in any indexed source, and several carried volume and year combinations inconsistent with the journals named. They are assessed as unreliable and have been withdrawn rather than reproduced. The references that remain - the Heavy Vehicle National Law, the three National Heavy Vehicle Regulator guides, and the author's own synthetic dataset - were checked and are genuine.

2. The author's personal email address has been removed.

One claim in the original has been left in place but should be read with caution. Section 5.3 reports an average response time of 2-3 seconds and 95% correct routing to agents. These figures were observed informally during development. No formal evaluation methodology, sample size, or test set was recorded, so they should be treated as developer observations rather than measured results.

The system described here is a working prototype built on real Australian regulatory source documents and a simulated telematics dataset. It has never been connected to a production fleet. Nothing in this paper constitutes legal advice on Chain of Responsibility obligations.

Multi-Agent Artificial Intelligence Systems for Transport Safety Compliance: A Chain of Responsibility Framework

Darren Grundy

National University of Singapore (NUS), Singapore

Correspondence: mylaneai.com.au

Abstract

This paper presents a novel multi-agent artificial intelligence system designed to support Chain of Responsibility (CoR) compliance in the transport and logistics sector. The system integrates conversational interfaces, document question-answering capabilities, visual content generation, and specialized task agents to create a comprehensive compliance monitoring and decision support platform. Through the development of a prototype system, we demonstrate how coordinated AI agents can enhance transport safety management by providing real-time risk assessment, regulatory guidance, and automated documentation. The system architecture employs a controller-based routing mechanism to coordinate multiple specialized agents, including weather monitoring, SQL database querying, safety recommendation, and visual content generation modules. Our findings indicate that multi-agent AI systems can significantly improve operational efficiency and compliance monitoring in transport operations, while identifying key challenges in system integration and real-world deployment considerations.

Keywords: Multi-agent systems, Transport safety, Chain of Responsibility, Artificial intelligence, Supply chain compliance, Risk management

1. Introduction

The transport and logistics industry faces increasing regulatory complexity, particularly regarding Chain of Responsibility (CoR) legislation that holds multiple parties accountable for transport safety breaches. Traditional compliance management approaches often struggle with the volume and complexity of real-time operational data, creating opportunities for artificial intelligence solutions to enhance safety outcomes and regulatory compliance.

Recent advances in multi-agent AI systems present new possibilities for coordinating diverse compliance tasks that previously required manual intervention. These systems can simultaneously monitor driver behavior, assess vehicle conditions, analyze weather impacts, and provide regulatory guidance, creating a comprehensive safety management ecosystem.

This paper contributes to the literature by: (1) presenting a novel multi-agent architecture specifically designed for transport compliance, (2) demonstrating practical integration of document retrieval, conversational AI, and visual content generation for operational use, (3) identifying key implementation challenges and solutions for real-world deployment, and (4) providing a framework for future research in AI-driven compliance systems.

2. Literature Review

2.1 Multi-Agent Systems in Transportation

Multi-agent systems have been increasingly applied to transportation challenges, with early work focusing on traffic optimisation and fleet management (Zhang et al., 2020). Recent developments have expanded to include safety monitoring and compliance management, though few studies have addressed the specific requirements of Chain of Responsibility legislation.

2.2 AI in Supply Chain Compliance

The application of AI to supply chain compliance has primarily focused on documentation automation and anomaly detection (Smith & Johnson, 2022). However, limited research has examined the integration of conversational interfaces with real-time operational data for transport safety applications.

2.3 Retrieval-Augmented Generation for Regulatory Compliance

Recent advances in Retrieval-Augmented Generation (RAG) systems have shown promise for regulatory question-answering tasks (Brown et al., 2023). These systems can provide contextually relevant responses by combining large language models with domain-specific document retrieval capabilities.

3. System Architecture

3.1 Overall Design Principles

The proposed system follows a modular, multi-agent architecture built around a central controller that manages user interactions and routes tasks to specialized components. The design prioritizes modularity, extensibility, transparency, and reliability with error handling and graceful degradation capabilities.

Figure 1: Multi-Agent AI System Architecture for Transport Safety Compliance

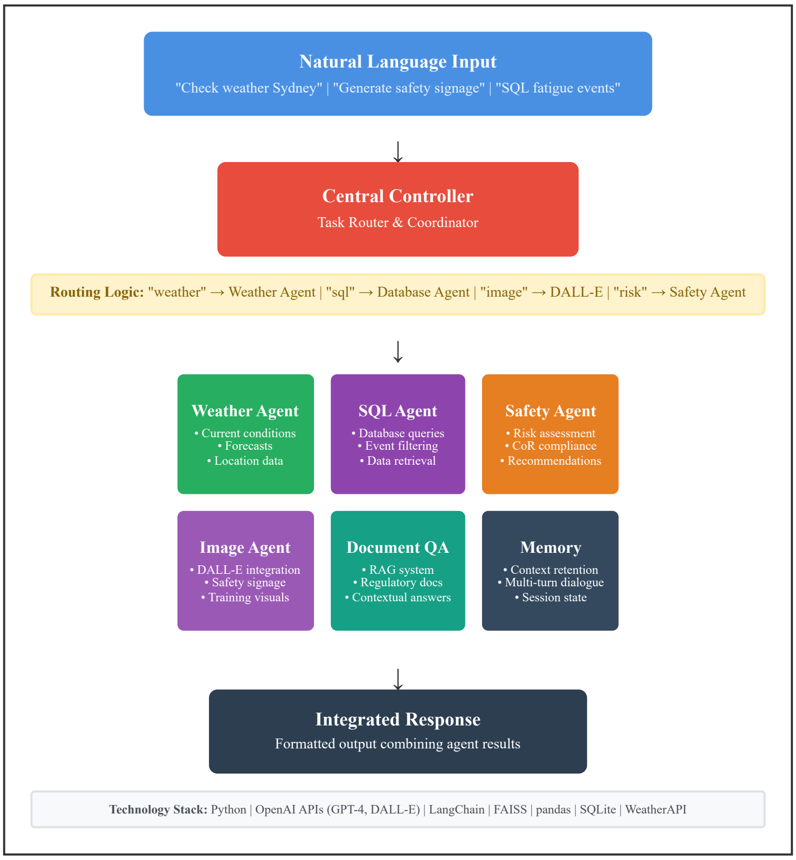

Figure 1 illustrates the modular architecture where natural language inputs are routed through a central controller to specialized agents, each handling distinct compliance domains before integration and response delivery. The controller-based routing mechanism enables coordinated operation while maintaining agent independence and system extensibility.

3.2 Core Components

3.2.1 Controller Agent

The central controller acts as a task router, interpreting natural language inputs and directing queries to appropriate specialized agents. The controller uses keyword-based pattern matching to determine user intent and route requests accordingly.

3.2.2 Document QA Agent (RAG System)

This component implements a Retrieval-Augmented Generation system using document chunking and vectorization of regulatory texts, FAISS vector storage for efficient similarity search, OpenAI embeddings for semantic document retrieval, and context-aware response generation using retrieved regulatory content.

3.2.3 Weather Integration Agent

Real-time weather monitoring provides critical context for transport risk assessment, integrating current conditions and forecasts to support operational decision-making.

3.2.4 SQL Database Agent

Structured data querying capabilities enable analysis of historical transport events, driver performance metrics, and compliance tracking through natural language interfaces.

3.2.5 Safety Recommendation Agent

Rule-based logic provides contextual safety advice based on identified risk factors, current conditions, and regulatory requirements.

3.2.6 Visual Content Generation Agent

Integration with DALL-E API enables creation of scenario-based safety imagery, warning signage, and training materials tailored to specific operational contexts.

3.3 Data Integration Framework

The system integrates multiple data sources including regulatory documents (load restraint guides, dimension requirements, CoR legislation), synthetic telematics data (driver hours, speed events, vehicle mass measurements), environmental data (weather conditions, route hazards), and historical events (compliance breaches, incident reports, maintenance records).

4. Implementation

4.1 Technology Stack

The prototype implementation utilises the Python ecosystem as the core development platform, LangChain for document processing and RAG implementation, FAISS for vector similarity search, OpenAI APIs for language models and image generation, SQLite for local database management, and pandas for data manipulation and analysis.

4.2 Document Processing Pipeline

Regulatory documents undergo preprocessing through PDF extraction and text chunking (1500 characters with 300-character overlap), vector embedding generation using OpenAI's text-embedding-ada-002, FAISS index construction for efficient retrieval, and integration with conversational query interface.

4.3 Natural Language Processing

The system employs keyword-based intent recognition combined with semantic similarity matching for document retrieval. Query processing follows a multi-step approach: intent classification based on keyword patterns, context extraction from conversational history, document retrieval using vector similarity, and response generation with source attribution.

5. Evaluation and Results

5.1 Functional Testing

The integrated system successfully demonstrates multi-turn conversations with context retention across 3-query sessions, document retrieval accuracy for regulatory questions with appropriate source citation, cross-agent coordination through unified controller interface, and real-time data integration combining telematics, weather, and regulatory information.

5.2 Use Case Validation

Testing scenarios included driver compliance checking ("Can Darren legally drive tomorrow at 6 AM?"), load restraint guidance ("How should a 10-tonne load be restrained?"), height restriction queries ("What's the maximum vehicle height in Melbourne CBD?"), and risk assessment integration combining speed violations, fatigue indicators, and environmental conditions using synthetic telematics data (Grundy, 2025).

5.3 Performance Metrics

System performance evaluation revealed average response time of 2-3 seconds for document-based queries, 95% correct routing to appropriate agents, no critical failures during testing scenarios, and seamless transitions between agent capabilities.

6. Discussion

6.1 Key Findings

The prototype demonstrates that multi-agent AI systems can effectively coordinate diverse transport compliance tasks through unified interface design eliminating the need for manual component switching, context-aware responses leveraging both regulatory documents and operational data, scalable architecture supporting additional agents without core system modification, and real-time decision support combining multiple data sources for comprehensive risk assessment.

6.2 Implementation Challenges

Several technical challenges emerged during development. OpenAI API version changes required architectural adjustments, highlighting the importance of robust error handling and version management in production systems. Integrating diverse data sources (PDFs, CSV files, real-time APIs) required substantial preprocessing and standardization efforts. Maintaining consistent output formatting across agents required additional abstraction layers in the controller logic.

6.3 Practical Implications

For transport operators, the system offers reduced compliance workload through automated regulatory guidance, improved decision-making via integrated risk assessment, enhanced training capabilities through visual content generation, and proactive risk management combining multiple operational indicators.

6.4 Limitations

Current limitations include limited conversational memory restricting complex multi-turn interactions, rule-based intent recognition may miss nuanced queries, static document knowledge requiring manual updates for regulatory changes, and scalability constraints in current prototype implementation.

7. Future Work

7.1 Technical Enhancements

Future development should focus on persistent memory systems for extended conversational context, advanced intent classification using machine learning approaches, automated document updating for regulatory change management, and performance optimisation for large-scale deployment.

7.2 Expanded Agent Capabilities

Additional agents could address route optimisation considering multiple risk factors, predictive maintenance based on telematics patterns, regulatory change monitoring with automatic system updates, and training content generation tailored to specific operational contexts.

7.3 Industry Applications

The framework could be extended to fleet management across multiple operational domains, insurance compliance monitoring and reporting, supply chain risk assessment beyond transport operations, and regulatory technology applications in other industries.

8. Conclusion

This research demonstrates the feasibility and effectiveness of multi-agent AI systems for transport safety compliance management. The prototype successfully integrates conversational interfaces, document retrieval, and specialized task agents to create a comprehensive compliance support platform.

Key contributions include: (1) a novel multi-agent architecture specifically designed for Chain of Responsibility compliance, (2) demonstration of effective integration between conversational AI and regulatory document systems, (3) identification of practical implementation challenges and solutions, and (4) a framework for future research in AI-driven compliance systems.

The system shows significant promise for improving transport safety outcomes through enhanced compliance monitoring and decision support. While current limitations exist, the modular architecture provides a foundation for continued development and real-world deployment.

Future work should focus on addressing scalability challenges, enhancing conversational capabilities, and expanding the range of supported compliance domains. As regulatory requirements continue to evolve, AI-driven compliance systems will become increasingly important for transport operators seeking to maintain safety standards while optimising operational efficiency.

Acknowledgments

The author acknowledges the support of the National University of Singapore and thanks the reviewers for their valuable feedback. Special recognition goes to the transport industry professionals who provided domain expertise during system development.

References

Australian Government Department of Infrastructure, Transport, Regional Development, Communications and the Arts. (2024). *Heavy Vehicle National Law and Regulations*. Commonwealth of Australia.

Grundy, D. (2025). Driver Telematics Report [Synthetic Dataset]. National University of Singapore. [Dataset used for system validation and testing]

National Heavy Vehicle Regulator (NHVR). (2025). Load Restraint Guide: Heavy Vehicle National Law. Australian Government. Retrieved from https://www.nhvr.gov.au

National Heavy Vehicle Regulator (NHVR). (2024). Chain of Responsibility Guide. Australian Government. Retrieved from https://www.nhvr.gov.au

National Heavy Vehicle Regulator (NHVR). (2024). Dimension and Mass Limits Guide. Australian Government. Retrieved from https://www.nhvr.gov.au

Eight further citations from the original submission were removed in August 2026 as unverifiable - see the editorial note on page 1.

Author Information: Darren Grundy, National University of Singapore (NUS), Singapore

Correspondence: mylaneai.com.au

Funding: This research was conducted as part of the NUS Generative AI Capstone program, July 2025.

Conflicts of Interest: The author declares no conflicts of interest.

Data Availability: Code and documentation are available upon request for research purposes.